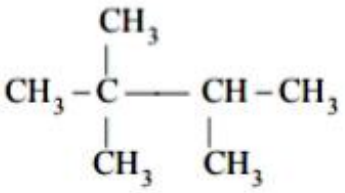
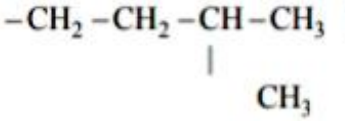
ALKANES

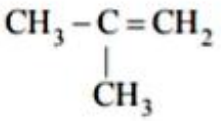

Triptane


Isopentyl Group

ALKENES

$CH_3–CH_2–CH=CH_2$ α–Butylene

$CH_3–CH=CH–CH_3$ β–Butylene


Iso Butylene

ALKYNES

$HC\equiv CH$ Purified Acetylene or Norcelyne

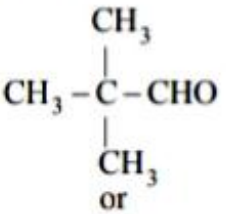
$CH_3–C\equiv CH$ Allylene

ETHER

$CH_3CH(OCH_3)_2$ Methylal

ALDEHYDE

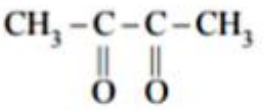

Glyoxalic acid

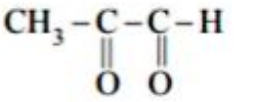

Pivaldehyde

or

$(CH_3)_3C–CHO$

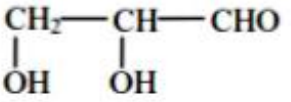
$(CH_3)_2CHCHO$ Isobutyraldehyde


Dimethyl Glyoxal


Methyl Glyoxal or
Pyruvialdehyde

$CH_3–CH=CH–CHO$ Crotonaldehyde

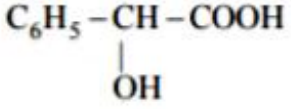
$CH_2=CH–CHO$ Acraldehyde (Acrolein)


Glyceraldehyde

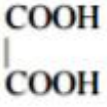

Glyoxal

CARBOXYLIC ACID

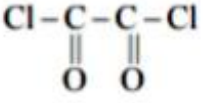
$CH_3–CO–COOH$ Pyruvic Acid


Mendalic Acid

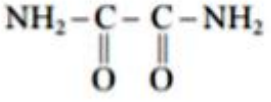
NH_2COOH Carbamic Acid
(Amino formic Acid)

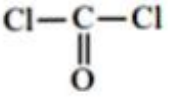

Oxalic acid

ACID DERIVATIVES

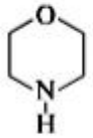

Oxalyl Chloride

NH_2COONH_4 Ammonium Carbamate


Oxanamide


Phosgene

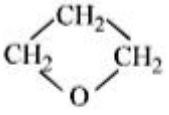
HETROCYCLIC COMPOUNDS



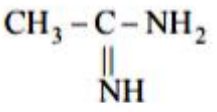
Morpholine



Hexa-methylenetetramine or Urotropine



Oxitane



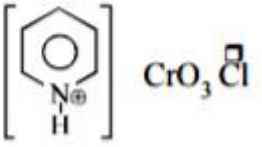
Amidine

SOME REAGENTS

LAH — Lithium aluminium hydride : $LiAlH_4$

SBH — Sodium borohydride $NaBH_4$

PCC — Pyridinium chlorochromate

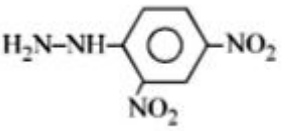


Raney Nickel — Ni–Al alloy

Wilkinson's Tris(Triphenylphosphine) catalyst — chlororhodium (I) $(PPh_3)_3RH^{\oplus}Cl^{\ominus}$

Bayer's reagent — 1% dil. alkaline aq.sol. of $KMnO_4$

Braddy's reagent 2,4 DNP



Liemieux reagent $NaIO_4$ + dil.alk.$KMnO_4$

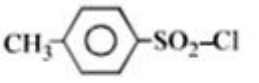
TEL — Tetra ethyl lead

Gillman's reagent — $R_2CuLi/[R_2Cu]^- Li^+$

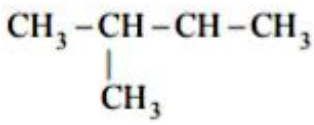
Tollen's reagent — alk. sol. of $AgNO_3$

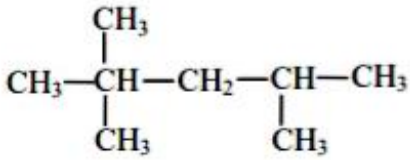
Fehling's reagent — alk. sol. of $CuSO_4$

Hinsberg's reagent

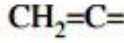


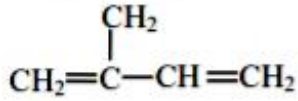
ALKANE

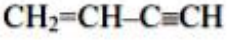

Isopentane


Isooctane

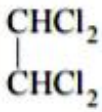
ALKENE

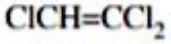

Allene


Isoprene

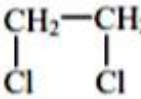

Vinyl acetylene

ALKYL HALIDE


Westron (Solvent)


Westrosol or Triclean (Solvent)


Ethylidene chloride (gem dihalide)

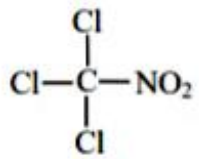

Ethylene Dichloride (Vinyl dihalide)

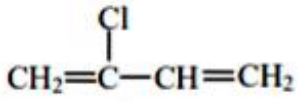

Methylene chloride


Chloroform

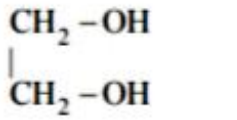

Carbontetra chloride

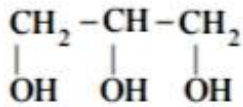
Tetraclean or Perclean

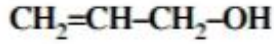

Chloropicrin (Tear gas)

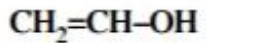

Chloroprene

ALCOHOL

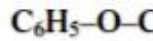

Glycol or Ethylene Glycol


Glycerol


Allyl Alcohol


Vinyl Alcohol

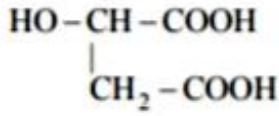
ETHER


Anisole
(Methyl Phenyl Ether)

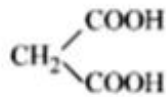
KETONE

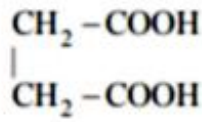

Acetone

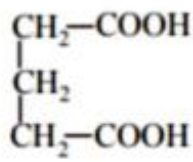
CARBOXYLIC ACID

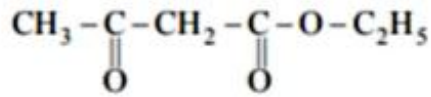

Malic acid

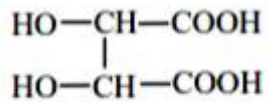

Glycolic Acid


Malonic acid

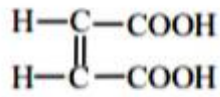

Succinic acid


Glutaric acid

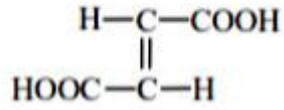

Aceto Acetic Ester **(AAE)**
or Ethyl Aceto Acetate



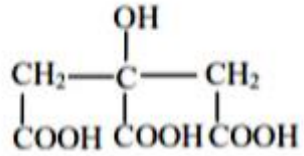
Tartaric acid



Maleic acid



Fumaric acid



Citric acid

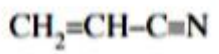


Crotonic acid

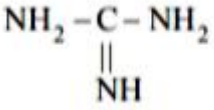


Cinnamic acid

N–DERIVATIVES



Vinyl Cyanide or Acrylo Nitrile

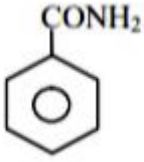


Guanidine

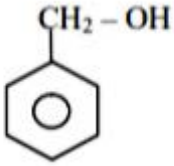
AROMATIC COMPOUNDS



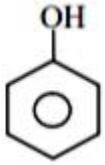
Benzene



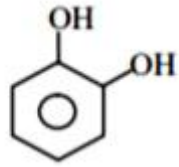
Benzamide



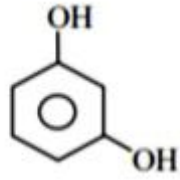
Benzyl alcohol (Aliphatic alcohol)



Phenol (Aromatic alcohol)



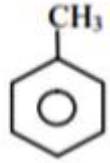
Catechol



Resorcinol



Quinol



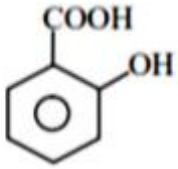
Toluene



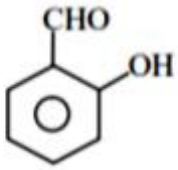
Pyrrole



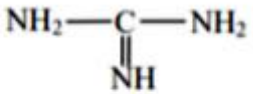
Furan



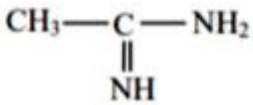
Salicylic acid



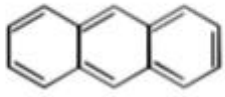
Salicylaldehyde



Guanidine



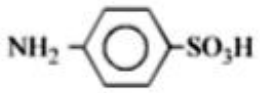
Amidine



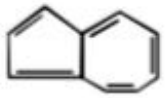
Anthracene



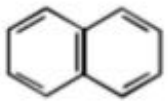
Thiophene



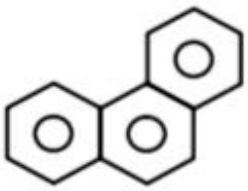
Sulphanilic acid



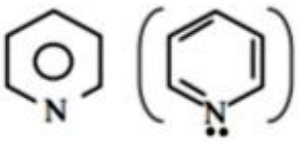
Azulene



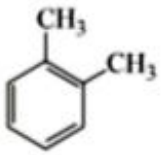
Napthalene



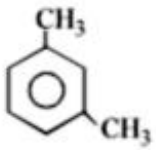
Phenanthrene



Pyridine



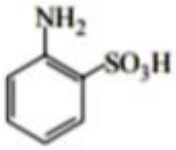
o–xylene



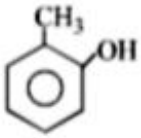
m-xylene



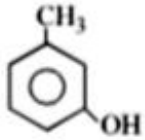
Nitrobenzene (oil of mirbane)



Orthanilic Acid

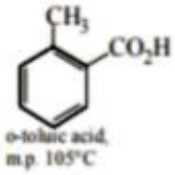


o-Cresol

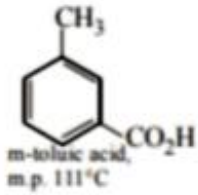


m-Cresol

$C_6H_5CO_3H$ Perbenzoic acid



o-toluic acid, m.p. 105°C

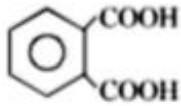


m-toluic acid, m.p. 111°C

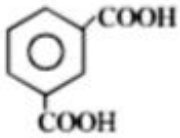


p-toluic acid, m.p. 180°C

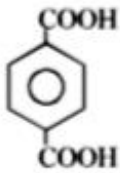
Toluic acids



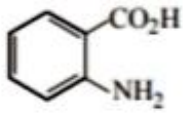
Phthalic acid



Isophthalic acid

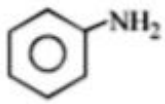


Terephthalic acid



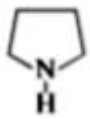
Anthranilic acid (o-aminobenzoic acid)

C_6H_5CHO Benzaldehyde

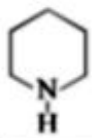


Aniline

HETROCYCLIC COMPOUNDS



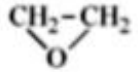
Pyrrolidine



Piperidine



Tetrahydrofuran (THF)



Oxirane or Ethylene Oxide or Oxo Cyclo Propane

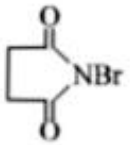


Quinuclidine

SOME REAGENTS

Grignard's reagent — RMgX

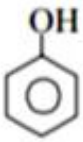
NBS — N-Bromosuccinimide



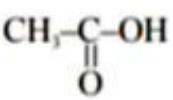
POLAR PROTIC SOLVENTS

H –O–H — Water

R–O–H — Alcohol



Phenol



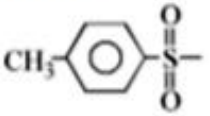
Acetic acid

HF — Hydrogen Fluoride

NH_3 — Ammonia

SOME GROUPS

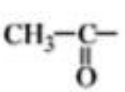
Ts — Tosyl



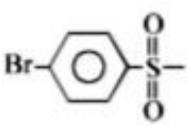
Ms — Mesyl



Ac — Acyl



Bs — Brosyl



Tf — Triflate



POLAR APROTIC SOLVENTS

DMS — Dimethyl sulphide $CH_3–S–CH_3$

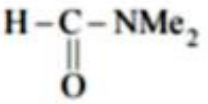
DMSO — Dimethyl sulphoxide $Me_2S=O$

HMPT — Hexamethylphosphoramide

or

HMPTA — $O=P–(NMe_2)_3$

DMF — Dimethyl formamide

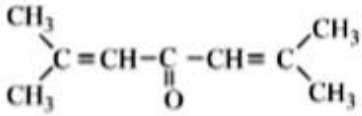


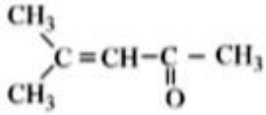
Crown ethers — Cyclic polyethers

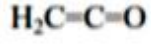


(12 – C – 4)

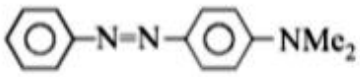
KETONE

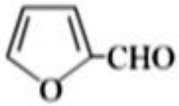

Phorone

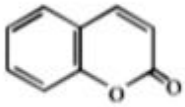

Mesityl Oxide

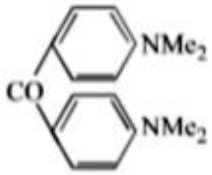

Ketene

AROMATIC COMPOUNDS


Butter Yellow

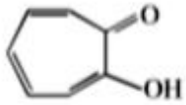

Furfural


Coumarine

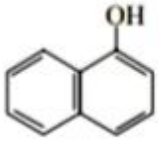

Michler's Ketone

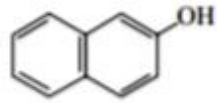

Tropone

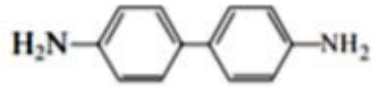
(Cycloheptatrienone)

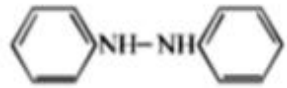

Tropolone

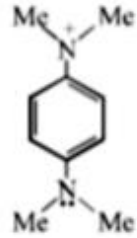
(Cycloheptatrienolone)

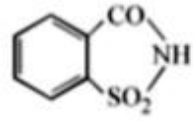

α–naphthol


β–naphthol


Benzidine

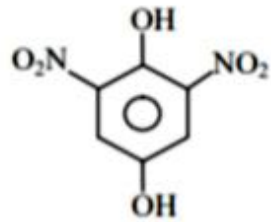

Hydrazobenzene


Wurster salts

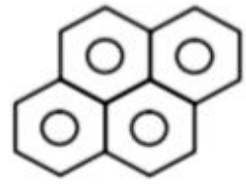

Saccharin (o-sulphobenzoic imide)

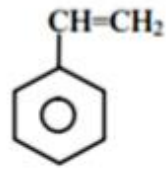
37 $C_6H_5COCOC_6H_5$ Benzil

$(C_6H_5)_2C(OH)CO_2H$ Benzilic acid


2, 4, 6 trinitrophenol (Picric acid)


Nitrobenzene (Oil of mirbane)


Pyrene


Styrene

Ph–CH=CH–Ph Stilbene